

ADVANCED PROBLEM PROGRAMING

SLO 1: Elements of Programming Languages

1. Syntax and Semantics of `int x = 5 + 3;`

Statement:

`int x = 5 + 3;`

Syntax:

Syntax refers to the rules for writing a valid Java statement.

- `int` → data type
- `x` → variable name/identifier
- `=` → assignment operator
- `5 + 3` → arithmetic expression
- `;` → statement terminator

The general syntax is:

`dataType variableName = expression;`

Semantics:

Semantics refers to the meaning or behavior of the statement.

Here:

1. `5 + 3` is calculated.
2. The result is 8.
3. The value 8 is assigned to the integer variable `x`.

So, after execution:

`x = 8;`

2. Identify Lexical Tokens in a Simple Java Program

Consider:

```
public class Test {  
    public static void main(String[] args) {  
        int x = 10;  
        System.out.println(x);  
    }  
}
```

The **lexical tokens** are the smallest meaningful units of a Java program.

Token	Example
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Keyword	public, class, static, void, int
---------	----------------------------------

Identifier	Test, main, args, x, System, out, println
------------	--

Literal	10
---------	----

Operator	=
----------	---

Separator	(,), {, }, [,], ;, .
-----------	------------------------

For example:

```
int x = 10;
```

contains:

- int → keyword
- x → identifier
- = → operator
- 10 → integer literal
- ; → separator

3. Modify a Sample Program to Demonstrate Grammar Rules in Java

Java grammar defines how keywords, identifiers, operators, expressions, and statements must be arranged.

Example:

```
class GrammarDemo {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 20;  
        int sum = a + b;  
  
        System.out.println("Sum = " + sum);  
    }  
}
```

This program demonstrates several grammar rules:

1. Class declaration

```
class GrammarDemo {  
}
```

2. Method declaration

```
public static void main(String[] args) {  
}
```

3. Variable declaration and initialization

```
int a = 10;
```

4. Expression

```
a + b
```

5. Assignment

```
int sum = a + b;
```

6. Method invocation

```
System.out.println("Sum = " + sum);
```

7. Statement termination

Each statement ends with ;.

Example of an incorrect grammar

```
int = 10 a;
```

This is invalid because the identifier must come after the data type:

```
int a = 10;
```